Real Time Video Empty Shelf Dector

Utilize trained YOLO11N and the videos I took in Taiwanese retail stores to detect empty shelves.

```
!nvidia_smi
```

```
/bin/bash: line 1: nvidia_smi: command not found
```

Download necessary libraries

```
!pip install ultralytics roboflow -q
import ultralytics
ultralytics.checks()
from ultralytics import YOLO
import os
import roboflow
from roboflow import Roboflow
import subprocess

# Mounting the videos and the model to this Colab
from google.colab import drive
drive.mount('/content/drive')
```

```
Ultralytics 8.4.80 🚀 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Setup complete ✅ (2 CPUs, 12.7 GB RAM, 47.3/112.6 GB disk)
Mounted at /content/drive
```

```
model = YOLO('/content/drive/MyDrive/shelf_project/best_v1.5.pt')


videos = [
    '/content/drive/MyDrive/1.MOV',
    '/content/drive/MyDrive/2.MOV',
    '/content/drive/MyDrive/3.MOV',
    '/content/drive/MyDrive/4.MOV',
    '/content/drive/MyDrive/5.MOV',
    '/content/drive/MyDrive/6.MOV'
]


for v in videos:
  model.track(
      source = v,
      conf = 0.10,
      save = True
  )
```

```
video 1/1 (frame 1031/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.8ms
video 1/1 (frame 1032/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 15.0ms
video 1/1 (frame 1033/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 11.2ms
video 1/1 (frame 1034/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.2ms
video 1/1 (frame 1035/1061) /content/drive/MyDrive/6.MOV: 384x640 4 empty shelf gaps, 13.9ms
video 1/1 (frame 1036/1061) /content/drive/MyDrive/6.MOV: 384x640 3 empty shelf gaps, 15.4ms
video 1/1 (frame 1037/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 15.3ms
video 1/1 (frame 1038/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.6ms
video 1/1 (frame 1039/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.0ms
video 1/1 (frame 1040/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.1ms
video 1/1 (frame 1041/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.9ms
video 1/1 (frame 1042/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.2ms
video 1/1 (frame 1043/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.9ms
video 1/1 (frame 1044/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.9ms
video 1/1 (frame 1045/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.6ms
video 1/1 (frame 1046/1061) /content/drive/MyDrive/6.MOV: 384x640 2 empty shelf gaps, 13.4ms
video 1/1 (frame 1047/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.4ms
video 1/1 (frame 1048/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.3ms
video 1/1 (frame 1049/1061) /content/drive/MyDrive/6.MOV: 384x640 2 empty shelf gaps, 12.6ms
video 1/1 (frame 1050/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 15.4ms
video 1/1 (frame 1051/1061) /content/drive/MyDrive/6.MOV: 384x640 2 empty shelf gaps, 13.8ms
video 1/1 (frame 1052/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.2ms
video 1/1 (frame 1053/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.4ms
video 1/1 (frame 1054/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.7ms
video 1/1 (frame 1055/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 13.7ms
video 1/1 (frame 1056/1061) /content/drive/MyDrive/6.MOV: 384x640 2 empty shelf gaps, 13.7ms
video 1/1 (frame 1057/1061) /content/drive/MyDrive/6.MOV: 384x640 2 empty shelf gaps, 16.7ms
video 1/1 (frame 1058/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 14.0ms
video 1/1 (frame 1059/1061) /content/drive/MyDrive/6.MOV: 384x640 1 empty shelf gap, 12.9ms
video 1/1 (frame 1060/1061) /content/drive/MyDrive/6.MOV: 384x640 2 empty shelf gaps, 9.0ms
video 1/1 (frame 1061/1061) /content/drive/MyDrive/6.MOV: 384x640 2 empty shelf gaps, 8.8ms
Speed: 4.1ms preprocess, 16.1ms inference, 1.7ms postprocess per image at shape (1, 3, 384, 640)
```

Try it with BoT-SORT with re-ID enabled

```
model.track(
    source = '/content/drive/MyDrive/4.MOV',
    conf = 0.10,
    save = True,
    tracker = 'botsort.yaml',
    )
```

```
        [  0,   0,   0],
        [  0,   0,   0],
        ...,
        [  0,   0,   0],
        [  0,   0,   0],
        [  0,   0,   0]],

       [[  0,   0,   0],
        [  0,   0,   0],
        [  0,   0,   0],
        ...,
        [  0,   0,   0],
        [  0,   0,   0],
        [  0,   0,   0]],

       [[  0,   0,   0],
        [  0,   0,   0],
        [  0,   0,   0],
```

```
Start coding or generate with AI.
```